

Gebze Technical University

Department of Electronics Engineering

ELM367: Digital Signal Processing

Lecture 4: Frequency-Domain Representation of Discrete-Time Signals and Systems

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Outline

Frequency-Domain Repr. of DT Signals and Systems

Eigen Function Of LTI System

What is Eigenfunction?

Any non-zero function f that when acted upon a system $\mathcal{S}$ is only scaled by a factor s referred to as eigenvalue:

$$\mathcal{S}\{f\} = sf$$

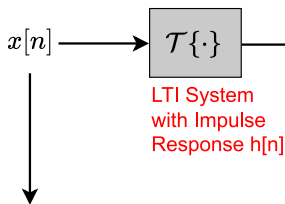
Example: $f = \sin(2x)$, $\mathcal{S} = \frac{d^2}{dx^2}$

$$\begin{aligned}\frac{df}{dx} &= \frac{d(\sin(2x))}{dx} = 2 \cos(2x) \\ \frac{d^2f}{dx^2} &= \frac{d(2 \cos(2x))}{dx} = -4 \cdot \underbrace{\sin(2x)}_{=f}\end{aligned}$$

$\therefore \sin(2x)$ is an eigenfunction to the system $\frac{d^2}{dx^2}$ with eigenvalue $= -4$

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Eigen Function Of LTI System



$$x[n] = Ae^{j(\omega n + \theta)}$$

∴ Complex Exponential are
Eigenfunctions of LTI systems

$H(e^{j\omega})$ is their Eigenvalue

$$\begin{aligned} y[n] &= \mathcal{T}\{Ae^{j(\omega n + \theta)}\} \\ &= h[n] * x[n] \\ &= \sum_{k=-\infty}^{\infty} h[k]x[n-k] \\ &= \sum_{k=-\infty}^{\infty} h[k] \cdot Ae^{j(\omega(n-k) + \theta)} \\ &= \sum_{k=-\infty}^{\infty} h[k] \cdot Ae^{j(\omega n + \theta)} \cdot e^{-j(\omega k)} \\ &= \underbrace{Ae^{j(\omega n + \theta)}}_{\text{= Input}} \underbrace{\sum_{k=-\infty}^{\infty} h[k] \cdot e^{-j(\omega k)}}_{H(e^{j\omega})} \end{aligned}$$

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Frequency Response Of LTI System

$$y[n] = H(e^{j\omega}) \cdot A e^{j(\omega n + \theta)} \quad \text{where} \quad H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k}$$

$H(e^{j\omega}) \implies$ is referred to as **Frequency Response** of an LTI System

$H(e^{j\omega}) \implies$ is generally Complex valued and can be represented as:

$$H(e^{j\omega}) = \underbrace{H_R(e^{j\omega}) + jH_I(e^{j\omega})}_{\text{Cartesian form}} = \underbrace{|H_R(e^{j\omega})| e^{j\angle H_R(e^{j\omega})}}_{\text{Polar form}}$$

Relationship between Cartesian and Polar Forms

Magnitude/Gain: $|H_R(e^{j\omega})| = \sqrt{H_R(e^{j\omega})^2 + H_I(e^{j\omega})^2}$

Phase: $\angle H_R(e^{j\omega}) = \tan^{-1} \left(\frac{H_I(e^{j\omega})}{H_R(e^{j\omega})} \right)$

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Frequency Response Of LTI System

Example: What is the frequency response of an Ideal delay system
 $y[n] = x[n - n_o]$

Solution:

For a complex sinusoidal input $x[n] = e^{j\omega n}$

$$y[n] = e^{j\omega(n-n_o)} = \underbrace{e^{-j\omega n_o}}_{H(e^{j\omega})} \cdot \underbrace{e^{j\omega n}}_{\text{Input}}$$

Alternative Solution: For Ideal delay System: $h[n] = \delta[n - n_o]$:

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} = \sum_{k=-\infty}^{\infty} \delta[k - n_o]e^{-j\omega k} = e^{-j\omega n_o}$$

$|H(e^{j\omega})| = ?$, $\angle H(e^{j\omega}) = ?$

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Frequency Response Of LTI System

For input $x[n] = Ae^{j(\omega n + \theta)}$, the output is given as:

$$y[n] = H(e^{j\omega}) \cdot Ae^{j(\omega n + \theta)}$$

For input that is **linear combination of multiple sinusoids**, i.e.,

$$x[n] = \sum_i A_i e^{j(\omega_i n + \theta_i)}$$

The output is

$$y[n] = \sum_i H(e^{j\omega_i}) \cdot A_i e^{j(\omega_i n + \theta_i)}$$

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Frequency Response Of LTI System

Example: Find $y[n]$ for a sinusoidal input $x[n] = B \cos(\omega_o n + \phi)$

Solution

$$\begin{aligned}x[n] &= B \cos(\omega_o n + \phi) \\&= \frac{B}{2} e^{j(\omega_o n + \phi)} + \frac{B}{2} e^{-j(\omega_o n + \phi)} \\&= \underbrace{\frac{B}{2} e^{j\phi} e^{j\omega_o n}}_{x_1[n]} + \underbrace{\frac{B}{2} e^{-j\phi} e^{-j\omega_o n}}_{x_2[n]}\end{aligned}$$

$$\begin{aligned}y[n] &= y_1[n] + y_2[n] \\&= H(e^{j\omega_o}) \cdot \frac{B}{2} e^{j\phi} e^{j\omega_o n} + H(e^{-j\omega_o}) \cdot \frac{B}{2} e^{-j\phi} e^{-j\omega_o n}\end{aligned}$$

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Periodicity of Frequency Response Of LTI System

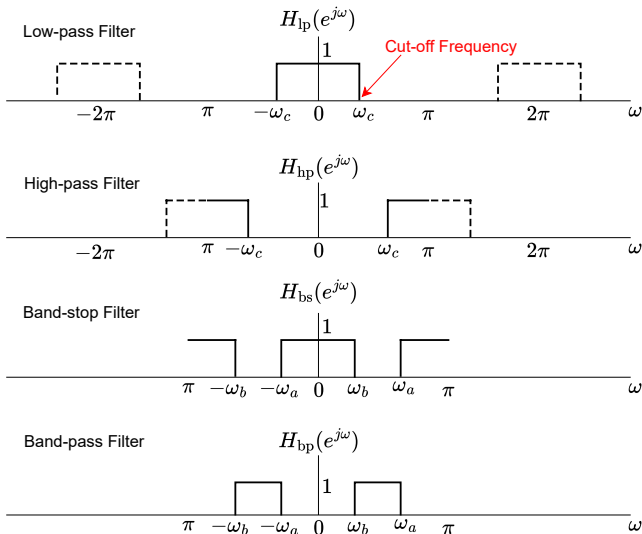
$H(e^{j\omega})$ is always periodic function of the frequency variable ω with period 2π .

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}$$

$$\begin{aligned} H(e^{j(\omega+2\pi)}) &= \sum_{k=-\infty}^{\infty} h[k]e^{-j(\omega+2\pi)k} \\ &= \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \underbrace{e^{-j2\pi k}}_{=1} \\ &= \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \\ &= H(e^{j\omega}) \end{aligned}$$

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Ideal Filters



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Moving Average System as a Filter

A moving average system is defined by:

$$y[n] = \frac{1}{M_1 + M_2 + 1} \sum_{k=-M_1}^{M_2} x[n - k]$$

What is its impulse response?

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Moving Average System as a Filter

A moving average system is defined by:

$$y[n] = \frac{1}{M_1 + M_2 + 1} \sum_{k=-M_1}^{M_2} x[n - k]$$

To find the impulse response, $h[n]$ we set $x[n] = \delta[n]$

$$h[n] = \frac{1}{M_1 + M_2 + 1} \sum_{k=-M_1}^{M_2} \delta[n - k] = \frac{1}{M_1 + M_2 + 1}$$

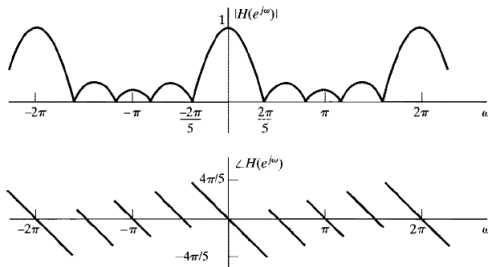
$$h[n] = \begin{cases} \frac{1}{M_1 + M_2 + 1} & , -M_1 \leq n \leq M_2 \\ 0 & , \text{Otherwise} \end{cases}$$

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Moving Average System as a Filter

$$\begin{aligned} H(e^{j\omega}) &= \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} = \sum_{k=-M_1}^{M_2} \frac{1}{M_1 + M_2 + 1} e^{-j\omega k} \\ &= \frac{1}{M_1 + M_2 + 1} \sum_{k=-M_1}^{M_2} e^{-j\omega k} = \frac{1}{M_1 + M_2 + 1} \times \frac{e^{j\omega M_1} - e^{-j\omega(M_2+1)}}{1 - e^{-j\omega}} \\ &= \frac{1}{M_1 + M_2 + 1} \times \frac{\sin[\omega(M_1 + M_2 + 1)/2]}{\sin[\omega/2]} \times e^{j\omega(M_1 - M_2)} \end{aligned}$$

For $M_1 = 0$, and $M_2 = 4$



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Moving Average System as a Filter

Example-1: Given $y[n] = \frac{1}{2}x[n] + \frac{1}{2}x[n-1]$. Find $H(e^{j\omega})$ and plot its Magnitude response

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Moving Average System as a Filter

Example-2: Given $y[n] = \frac{1}{2}x[n] - \frac{1}{2}x[n-1]$. Find $H(e^{j\omega})$ and plot its Magnitude response

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Frequency Response of a Suddenly Applied Exponential Input

Recall: For input $x[n] = Ae^{j(\omega n + \theta)}$ to an LTI system with Impulse response $h[n]$, where $-\infty < n < \infty$, we had:

$$y[n] = H(e^{j\omega}) \cdot Ae^{j(\omega n + \theta)}, \quad \text{where} \quad H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}$$

Now: Let the input be $x[n] = e^{j\omega n}u[n - N]$ to a causal LTI system with Impulse response $h[n]$. What is the output $y[n]$?

$$y[n] = \begin{cases} e^{j\omega n} \sum_{k=0}^n h[k]e^{-j\omega k} & , n \geq 0 \\ 0 & , n < 0 \end{cases}$$

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Frequency Response of a Suddenly Applied Exponential Input

For $n \geq 0$

$$\begin{aligned}y[n] &= e^{j\omega n} \sum_{k=0}^n h[k] e^{-j\omega k} \\&= e^{j\omega n} \underbrace{\sum_{k=0}^{\infty} h[k] e^{-j\omega k}}_{=H(e^{j\omega})} - e^{j\omega n} \sum_{k=n+1}^{\infty} h[k] e^{-j\omega k} \\&= \underbrace{H(e^{j\omega}) e^{j\omega n}}_{\text{Steady-state response } y_{ss}[n]} - \underbrace{\left(\sum_{k=n+1}^{\infty} h[k] e^{-j\omega k} \right) e^{j\omega n}}_{\text{Transient response } y_{tr}[n]}\end{aligned}$$

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Frequency Response of a Suddenly Applied Exponential Input

Example-3: Given a system $y[n] = \frac{1}{3} (x[n] + x[n-1] + x[n-2])$.
Plot $y[n]$ when the input is $x[n] = u[n]$.

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Frequency Response of a Suddenly Applied Exponential Input

$$y[n] = \underbrace{H(e^{j\omega})e^{j\omega n}}_{\text{Steady-state response } y_{ss}[n]} - \underbrace{\left(\sum_{k=n+1}^{\infty} h[k]e^{-j\omega k} \right) e^{j\omega n}}_{\text{Transient response } y_{tr}[n]}$$

- ▶ In some cases, the transient response goes to zero after some time, i.e., $y_{tr}[n] \rightarrow 0$, and the system attains its steady state, i.e., $y[n] \rightarrow y_{ss}[n]$
- ▶ Let's analyze the magnitude of transient response $y_{tr}[n]$

$$|y[n]| = \left| \sum_{k=n+1}^{\infty} h[k]e^{-j\omega k} e^{j\omega n} \right| \leq \sum_{k=n+1}^{\infty} |h[k]|$$

For FIR Systems: $h[n] = 0$ except for $0 \leq n \leq M$

$$|y_{tr}[n]| = 0 \quad \text{for } n+1 > M \implies n > M-1.$$

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Frequency Response of a Suddenly Applied Exponential Input

For IIR Systems:

- ▶ $y_{tr}[n]$ does not disappear abruptly
- ▶ If $h[n] \rightarrow 0$ as $n \rightarrow \infty$,

$$|y_{tr}[n]| = \left| \sum_{k=n+1}^{\infty} h[k] e^{-j\omega k} e^{j\omega n} \right| \leq \sum_{k=n+1}^{\infty} |h[k]| \leq \underbrace{\sum_{k=0}^{\infty} |h[k]|}_{\text{Stable system}} < \infty$$

- ▶ For a **stable LTI system**, $|y_{tr}[n]| \rightarrow 0$ as $n \rightarrow \infty$
- ▶ **Stability is a sufficient condition** for y_{tr} to decay asymptotically

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Existence of Frequency Response $H(e^{j\omega})$

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}$$

- ▶ This sum might not always converge.
- ▶ It only converges when $h[k]$ is absolutely summable:

$$|H(e^{j\omega})| = \left| \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \right| = \sum_{k=-\infty}^{\infty} |h[k]e^{-j\omega k}| = \sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

- ▶ This is the sufficient condition for the existence of $H(e^{j\omega})$
- ▶ **FIR Systems:** absolutely summable $\implies$ Stable $\implies$ their $H(e^{j\omega})$ exist.
- ▶ **IIR Systems:** We must check the convergence of the infinite sum.

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Existence of Frequency Response $H(e^{j\omega})$

Example Given an LTI system $y[n] = \alpha y[n-1] + x[n]$ for $n \geq 0$ with initial condition $y[-1] = 0$. Find $H(e^{j\omega})$.